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ADVANCED DATABASE WK7

1.Scenario Analysis: Parallel Database Setup for an E-commerce Platform

An e-commerce platform uses a parallel database system to manage large amounts of data efficiently. The system distributes database tasks across multiple servers to improve performance and reduce processing time. For example, when customers browse products, place orders or make payments, the parallel database processes multiple requests simultaneously.

The database uses parallel processing to divide large tasks into smaller operations that can be handled by different processors. This allows the platform to serve many customers at the same time without significant delays. Data is distributed across multiple servers to balance workloads and improve response times. If one server fails, data replication can help maintain availability and reduce the risk of data loss.

The parallel database also supports faster searching, inventory management and order processing. By using this setup, the e-commerce platform can handle increased traffic, improve customer experience and support business growth while maintaining reliable database operations.

2. Concept Research: Shared-Nothing Architecture

Shared-nothing architecture is a distributed computing system in which each server has its own processor, memory and storage. Servers operate independently and communicate through a network to perform tasks. This architecture is commonly used in parallel databases, cloud computing and large-scale applications. Each server processes a portion of the workload, allowing multiple tasks to run simultaneously. Shared-nothing systems improve performance, scalability and reliability because additional servers can be added as demand increases. They also reduce resource competition since servers do not share memory or storage. However, data distribution and communication between servers must be carefully managed. This architecture is useful for businesses that handle large amounts of data and require fast, efficient processing.

3. Tool Practice: Simulating a Parallel Query in PostgreSQL

I simulated a parallel query in PostgreSQL to understand how parallel processing improves database performance. I created a sample table containing customer records and executed a query to retrieve and analyse the data. I enabled parallel query settings and used the EXPLAIN command to examine the query execution plan. The results demonstrated how PostgreSQL can divide certain queries into smaller tasks and process them using multiple workers. This practice helped me understand how parallel queries can reduce execution time when working with large datasets. I also learned that parallel processing depends on system resources, query complexity and database configuration. Overall, the exercise improved my understanding of database performance optimisation.

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